

# GenAI for Patient-Centric Healthcare

## Facilitator Handbook

Delivery notes, medical demonstration boundaries, lab runbooks, fallback routes, verification guidance, and answer key.

|                                                                       |                                                                           |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>60 min</b><br>Concise theory and medical demonstrations            | <b>90 min</b><br>Six case-based labs with verification                    |
| <b>4 pillars</b><br>Patient • Clinical • Pharma/Research • Governance | <b>Safety</b><br>Synthetic data • bounded task • qualified human decision |

**Important:** Every case, image, name, and value in this kit is synthetic. The materials are for education, not diagnosis, prescribing, treatment, regulatory submission, or autonomous decision-making.

# 1. Delivery at a glance

| Minutes | Slides / activity | Facilitator outcome                                                               |
|---------|-------------------|-----------------------------------------------------------------------------------|
| 00–08   | Slides 1–2        | Set the synthetic-data, bounded-task, source-check, and human-authority contract. |
| 08–22   | Slides 3–4        | Pillar 1 and synthetic prescription demonstration.                                |
| 22–36   | Slides 5–6        | Patient action plan and medical-image observation boundary.                       |
| 36–50   | Slides 7–8        | Clinical reasoning, PV intake, and evidence verification.                         |
| 50–60   | Slides 9–10       | Risk screen, tool selection, and lab assignment.                                  |
| 60–105  | Labs 1–3          | Prescription; image observation; discharge + multilingual teach-back.             |
| 105–150 | Labs 4–6          | Differential/red flags; PV intake; evidence/citation verification.                |

## Non-negotiable opening

Do not ask participants to upload real patient data, real prescriptions, scans, company-confidential documents, or identifiable case details. Use the included synthetic assets.

## 2. Four-pillar speaking notes

### 01 Patient Understanding and Engagement

Explain, translate, teach back.

Emphasise comprehension, teach-back, translation review, and exact medication/source fidelity.

### 02 Clinical Workflow and Decision Support

Structure, observe, escalate.

Emphasise adequacy, missingness, red flags, uncertainty, bias, and clinician-owned diagnosis/treatment.

### 03 Pharma, Research and Professional Productivity

Search, extract, document.

Separate intake structuring and discovery from PV, medical, regulatory, screening, and interpretation decisions.

### 04 Responsible AI, Governance and Implementation

Classify, control, monitor.

Classify risk by data, impact, autonomy, and reach; require ownership, monitoring, incident route, and stop rule.

### 3. Live demo: synthetic prescription

| Step      | Facilitator action                                                          | Watch for                           |
|-----------|-----------------------------------------------------------------------------|-------------------------------------|
| 1 Prepare | Open `synthetic_prescription_01.png`; keep CSV source values hidden.        | No real prescription or identifier. |
| 2 Prompt  | Use exact extraction, [UNREADABLE], explanation, and verification sections. | Model guesses or infers indication. |
| 3 Reveal  | Compare every field against `prescription_cases.csv`.                       | Dose/frequency/duration drift.      |
| 4 Decide  | Ask a clinician/pharmacist role to accept, edit, reject, or escalate.       | Fluency substituted for fidelity.   |

**Boundary**

The demonstration is not OCR validation, prescribing, medication reconciliation, interaction checking, or clinical advice.

## 4. Live demo: synthetic medical image

| Do                                                        | Do not                                          |
|-----------------------------------------------------------|-------------------------------------------------|
| State: fully synthetic and no diagnostic ground truth.    | Ask for or reveal a diagnosis.                  |
| Assess image adequacy and limitations first.              | Claim general-purpose model validation.         |
| Use neutral visible observations and cannot-assess items. | Allow emergency clearance or treatment advice.  |
| Require radiologist/treating-clinician interpretation.    | Use a real medical image in an unapproved tool. |

Use the official product limitation page during the debrief: <https://help.openai.com/articles/8400551-chatgpt-image-inputs-faq>

### Offline fallback

Show the static synthetic image and ask teams to rewrite an unsafe diagnostic prompt into an adequacy–observation–uncertainty–escalation prompt.

## 5. Demo stack and fallback routes

| Workflow         | Primary route         | Alternative                  | Offline fallback                  |
|------------------|-----------------------|------------------------------|-----------------------------------|
| Prescription     | ChatGPT               | Gemini / Microsoft Copilot   | Static image + exact source table |
| Medical image    | ChatGPT               | Gemini                       | Prompt-boundary critique          |
| PV intake        | ChatGPT / Claude      | Gemini / Julius AI           | Manual PV intake template         |
| Evidence         | PubMed / Elicit       | Consensus / Semantic Scholar | Synthetic abstract extraction     |
| Patient material | Canva + reviewed copy | Gamma                        | Workbook fact-sheet canvas        |

Product access, capabilities, regional availability, account eligibility, terms, and pricing can change. Verify on the official site before delivery.

## 6. Six lab runbooks

### Lab 1: Prescription extraction and patient explanation

| Field                  | Facilitator reference                                                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset                | prescription_cases.csv                                                                                                                                        |
| Primary / alternatives | ChatGPT / Google Gemini / Microsoft Copilot                                                                                                                   |
| Expected output        | Exact transcription table; plain-language explanation; ambiguity log; warning that the prescription itself remains authoritative; three teach-back questions. |
| Human reviewer         | clinician or registered pharmacist                                                                                                                            |
| Prohibited             | No prescribing, substitution, interaction decision, adherence judgement, or reconstruction of unreadable content.                                             |

Debrief: What changed from source? What could harm? What was verified? Who decided? What control and stop condition would implementation require?

### Lab 2: Structured medical-image observation

| Field                  | Facilitator reference                                                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset                | imaging_cases.csv                                                                                                                             |
| Primary / alternatives | ChatGPT / Google Gemini                                                                                                                       |
| Expected output        | Image adequacy checklist; structured visible observations; uncertainty; cannot-assess list; escalation and human-review note.                 |
| Human reviewer         | qualified radiologist or treating clinician                                                                                                   |
| Prohibited             | No diagnosis, triage clearance, treatment recommendation, or claim that a general-purpose model can read specialised medical images reliably. |

Debrief: What changed from source? What could harm? What was verified? Who decided? What control and stop condition would implementation require?

### Lab 3: Discharge plan with teach-back

| Field                  | Facilitator reference                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Dataset                | discharge_summaries.csv                                                                                                    |
| Primary / alternatives | ChatGPT / Claude / Google Gemini / Doximity Ask                                                                            |
| Expected output        | What happened; exact medicine/action schedule; red flags; follow-up; three teach-back questions; missing-information list. |
| Human reviewer         | treating clinician or discharge nurse                                                                                      |
| Prohibited             | No new clinical advice, dose change, diagnosis, or omission of escalation instructions.                                    |

Debrief: What changed from source? What could harm? What was verified? Who decided? What control and stop condition would implementation require?

### Lab 4: Differential and red-flag checklist

| Field                  | Facilitator reference                                                                                |
|------------------------|------------------------------------------------------------------------------------------------------|
| Dataset                | clinical_reasoning_cases.csv                                                                         |
| Primary / alternatives | OpenEvidence / Glass Health / Pathway / ChatGPT                                                      |
| Expected output        | Clinician-review differential checklist; cognitive-bias prompt; missing-data list; urgency triggers. |
| Human reviewer         | qualified treating clinician                                                                         |
| Prohibited             | No final diagnosis, treatment plan, prescription, or autonomous triage decision.                     |

Debrief: What changed from source? What could harm? What was verified? Who decided? What control and stop condition would implementation require?



## Lab 5: Pharmacovigilance intake structuring

| Field                  | Facilitator reference                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------|
| Dataset                | adverse_event_reports.csv                                                                                     |
| Primary / alternatives | ChatGPT / Claude / Google Gemini / Julius AI                                                                  |
| Expected output        | Neutral narrative; seriousness-screening rationale; missing fields; follow-up questions; escalation triggers. |
| Human reviewer         | qualified pharmacovigilance or safety physician reviewer                                                      |
| Prohibited             | No causality, expectedness, reportability, coding, or submission decision.                                    |

Debrief: What changed from source? What could harm? What was verified? Who decided? What control and stop condition would implementation require?

## Lab 6: Reproducible evidence search

| Field                  | Facilitator reference                                                                                    |
|------------------------|----------------------------------------------------------------------------------------------------------|
| Dataset                | research_questions.csv                                                                                   |
| Primary / alternatives | PubMed / Elicit / Consensus / Semantic Scholar / SciSpace                                                |
| Expected output        | Search protocol; screening log; evidence-table schema; citation-verification checklist; limitation note. |
| Human reviewer         | qualified researcher or medical reviewer                                                                 |
| Prohibited             | No invented paper, citation, evidence claim, or replacement for a systematic-review protocol.            |

Debrief: What changed from source? What could harm? What was verified? Who decided? What control and stop condition would implementation require?

## 7. Dropdown coverage and selection logic

| Selector         | Coverage rule                                                                      |
|------------------|------------------------------------------------------------------------------------|
| Pillar           | Options come from complete workflow records.                                       |
| Role             | Shows only roles remaining after pillar selection.                                 |
| Task             | Shows only tasks remaining after prior selections.                                 |
| Input / language | Unsupported intersections are removed before selection.                            |
| Free-text need   | Returns a clearly labelled closest verified alternative.                           |
| Tool / demo      | Every result includes dataset, prompt, output, verification, safety, and reviewer. |

Current catalogue: 24 complete workflows across four pillars. Automated tests enumerate every visible option and verify that it returns at least one workflow.

## 8. Assessment answer key

| Questions | Key concepts                                                                                                      |
|-----------|-------------------------------------------------------------------------------------------------------------------|
| 1–5       | Synthetic/approved data; professional accountability; mark unreadable; non-diagnostic observation; adequacy first |
| 6–10      | Preserve source; fluent clinical review; teach-back; checklist not authority; common vs cannot-miss               |
| 11–15     | Neutral PV intake; verify citations; reproducible logs; source checks; mark missing                               |
| 16–20     | Identifiable + autonomy raises risk; pilot controls; subgroup/language review; workflow fit; decision record      |

### Scoring guidance

80%+ strong foundation; 60–79% review the relevant explanation; below 60% revisit governance and verification before planning a pilot.

## 9. Troubleshooting and safety interventions

| Situation                           | Immediate response                                                                                                                              |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Participant starts using real data  | Stop, delete/close the input if possible, and return to included synthetic assets; follow institutional incident guidance if exposure occurred. |
| Model gives a diagnosis             | Do not debate correctness. Mark boundary breach; rewrite prompt; require qualified interpretation.                                              |
| Model fabricates a citation         | Open source, show failure, discard unsupported claim, and demonstrate citation verification.                                                    |
| No tool access / sign-in failure    | Use static source + manual prompt critique + expected-output and verification templates.                                                        |
| Dropdown search is unfamiliar       | Use the labelled closest verified alternative or cascade by pillar and role.                                                                    |
| Discussion becomes vendor promotion | Return to task, data boundary, evidence, reviewer, monitoring, and stop condition.                                                              |

## 10. Sources and verification log

Verified on 28 July 2026. Recheck before each delivery because product, policy, and regulatory information can change.

- **WHO: Ethics and governance of AI for health:** <https://www.who.int/publications/i/item/9789240029200>
- **NIST AI Risk Management Framework:** <https://www.nist.gov/itl/ai-risk-management-framework>
- **FDA: AI-enabled medical devices:**  
<https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-enabled-medical-devices>
- **FDA: Clinical decision support FAQ:**  
<https://www.fda.gov/medical-devices/software-medical-device-samd/clinical-decision-support-software-frequently-asked-questions-faqs>
- **ChatGPT image-input limitations:** <https://help.openai.com/articles/8400551-chatgpt-image-inputs-faq>
- **India Digital Personal Data Protection Act:**  
<https://www.meity.gov.in/static/uploads/2024/06/2bf1f0e9f04e6fb4f8fef35e82c42aa5.pdf>
- **India Digital Personal Data Protection Rules:** <https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWa>